

Operations Manual

OTS Manager CLI & Batch

Management of identities, groups, and QR Codes via REST API on OpenTAKServer

Document Control

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Objective. This manual presents a installation, configuration, operation, batch provisioning, QR Code generation, diagnostics, and best practices do utilitário `ots_manager.py`.

Table of Contents

1. Overview
2. Architecture and operational flow
3. Prerequisites and installation
4. Configuration via environment variables
5. Security and credential protection
6. REST API mapping
7. CLI commands
8. Batch provisioning
9. QR Codes and access validity
10. Operational procedures
11. Error handling and diagnostics
12. Best practices and limitations
13. Operations checklist
14. Quick reference
15. Appendix: code structure

1. Overview

OTS Manager is a Python utility for automating OpenTAKServer administration. It encapsulates the operations required to:

- authenticate to the backend via `/api/login`;
- maintain an HTTP session with cookies;
- retrieve and send CSRF tokens required by Flask-WTF/Flask-Security;
- send the `Referer` and `Origin` headers;
- create groups;
- create users with mandatory password confirmation;
- associate users with groups in the `IN` and `OUT` directions;
- generate QR Code strings and PNG images for Android and iPhone;
- process multiple users from a JSON file;
- consolidate the processing results into a JSON report.

1.1 Operating model

The recommended workflow is:

1. configure `OTS_URL`, `OTS_USER` e `OTS_PASS`;
2. validate connectivity with the server;
3. perform the login;
4. create the required groups;
5. create the user;
6. associate the user with groups in `IN` e `OUT`;
7. generate the QR Code for the correct ecosystem;
8. store the PNG and record the result;
9. validate activation on the ATAK/iTAK device.

1.2 Ecosystems suportados

Application	Endpoint	Operation	Result
Android / ATAK	<code>/api/atak_qr_string</code>	POST	QR Code configuration string
iPhone / iTAK	<code>/api/itak_qr_string</code>	GET	iOS configuration string

2. Architecture and operational flow

The script uses `requests.Session()` to preserve cookies and headers between calls. After login, the CSRF token can be obtained from the cookies `csrf_token`, `csrf_access_token` ou `XSRF-TOKEN`. If the server returns the token in the response body, the utility also attempts to retrieve it from the JSON.

2.1 Authentication flow

```
ots_manager.py
|
| POST /api/login
v
OpenTAKServer
|
| cookies de sessão + token CSRF
v
Sessão requests.Session()
|
+--> Referer / Origin
+--> X-CSRFToken / X-CSRF-TOKEN
+--> operações protegidas
```

2.2 Applied headers

The expected global headers are:

```
Content-Type: application/json
Referer: <OTS_URL>/
Origin: <OTS_URL>
```

Para protected operations, o token é enviado adicionalmente como:

```
X-CSRFToken: <token>
X-CSRF-TOKEN: <token>
```

3. Prerequisites and installation

3.1 Requirements

- Python 3.9 or higher;
- network access to OpenTAKServer;
- credentials for an account authorized to create users and groups;
- write permissions in the working directory;
- `pip` available in the Python environment.

3.2 Dependency installation

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install --upgrade pip
pip install requests qrcode pillow
```

The libraries have the following responsibilities:

Library	Function
requests	HTTP communication, session, and cookies
qrcode	QR Code image generation
Pillow	Image backend used by qrcode

3.3 Suggested organization

```
ots-manager/
├── ots_manager.py
├── usuarios.json
├── qrcodes/
├── resultado_qr_codes.json
└── .venv/
```

Do not version files containing passwords, tokens, QR strings, or real personal data.

4. Configuration via environment variables

Set the parameters before execution:

```
export OTS_URL="http://opentakserver.example.com:5000"
export OTS_USER="admin"
export OTS_PASS="sua_senha_aqui"
```

The script removes the trailing slash from `OTS_URL`. If the variables are absent, the default values are:

Variable	Code default	Recommendation
OTS_URL	http://localhost:5000	Always specify explicitly
OTS_USER	admin	Use a named or service account
OTS_PASS	admin_password	Never rely on the default

4.1 Connectivity test

```
curl -i "$OTS_URL/"
```

If the service requires HTTPS, use an `https://` and validate the certificate. Avoid disabling TLS validation in production.

4.2 Running without exposing the password in shell history

Whenever possible, provide the password through a secure environment mechanism. At a minimum, avoid placing the password directly in shared commands or versioned scripts:

```
read -r -s OTS_PASS
export OTS_PASS
python ots_manager.py --help
```

5. Security and credential protection

Warning: the original example uses HTTP and an IP address. In production, prefer HTTPS with a valid certificate, a restricted network, and an account with the least privilege necessary.

Best practices:

- do not store `OTS_PASS` no repositório;
- do not share `resultado_qr_codes.json` without removing QR strings;
- protect generated PNG files because they may enable device provisioning;
- use service accounts with password rotation;
- restrict server access using a firewall or VPN;
- record operations without recording passwords or tokens;
- validate the destination before running destructive or provisioning commands;
- remove temporary files containing credentials after use.

6. REST API mapping

Operation	Endpoint	Method	Main data	CSRF/Referer
Authentication	<code>/api/login</code>	POST	<code>username</code> , <code>password</code>	Sessão inicial
Create group	<code>/api/groups</code>	POST	<code>name</code>	Sim
Create user	<code>/api/user/add</code>	POST	<code>username</code> , <code>password</code> , <code>confirm_password</code> , <code>email</code> , <code>administrator</code>	According to OTS configuration
Link group	<code>/api/users/groups</code>	PUT	<code>username</code> , <code>groups[]</code> , <code>direction</code>	According to OTS configuration
QR Android	<code>/api/atak_qr_string</code>	POST	<code>username</code> , <code>exp</code> , <code>nb</code> , <code>max</code>	Sim
QR iPhone	<code>/api/itak_qr_string</code>	GET	defined by the server	Authenticated session

6.1 Required and optional fields

The table below consolidates the fields accepted pelo `ots_manager.py`. Fields marked as **opcionais** may be omitted; in which case the script uses the CLI default or lets OpenTAKServer apply its own policy.

Operation	Required fields	Optional fields and default behavior
Authentication — / api/login	username, password	None. They are obtained from OTS_USER e OTS_PASS ; o código possui valores padrão, mas recomenda-se configure ambos explicitamente.
Create group — / api/groups	name	Nenhum.
Create user — / api/user/add	username, password, confirm_password	email is not required and may be omitted or left empty; administrator is optional and defaults to false. O confirm_password é preenchido automaticamente pelo script com o mesmo valor de password.
Link group — /api/ users/groups	username, groups[] with at least one group, direction	Na CLI, direction is optional and defaults to BOTH , performing the IN e OUT .
QR Android — /api/ atak_qr_string	username	exp e max são opcionais . nbf is calculated automatically only when exp é informado. Se exp e max are omitted or null , are not sent and the server default policy applies.
QR iPhone — /api/ itak_qr_string	Authenticated session	Não há campos informados na requisição GET ; a configuração é retornada pelo servidor.
CLI create-user	--username, -- password	--email, --groups, --admin, --app, --exp, --max e --save-qr are optional. --app assume android; --admin defaults to false.
CLI qr	--username	--app, --exp, --max e --save-qr are optional. --app assume android.
CLI create-group	--name	Nenhum.
CLI link	--username, --group	--direction is optional and defaults to BOTH .
CLI batch	--file	--output is optional and defaults to resultado_qr_codes.json . Dentro de cada registro, username e password são obrigatórios; email, administrator, groups, app, expiration e max_uses are optional.

Summary of batch user fields

Field JSON	Required?	Notes
username	Sim	User identifier in OTS.
password	Sim	Initial password; the script also sends <code>confirm_password</code> com o mesmo valor.
email	Não	May be omitted, set to <code>null</code> or an empty string, depending on the OTS installation validation. The script sends it as empty when not provided.
administrator	Não	Assume <code>false</code> when omitted.
groups	Não	Group list; when provided, each group is associated in <code>IN</code> e <code>OUT</code> .
app	Não	Assume <code>android</code> ; also accepts <code>iphone</code> .
expiration	Não	Days or date <code>YYYY-MM-DD</code> ; when absent or <code>null</code> , no expiration is sent.
max_uses	Não	Limite de ativações; when absent or <code>null</code> , no limit is sent.

Important: `email`, `expiration` e `max_uses` are not required fields. The absence of these fields does not prevent user creation or QR Code generation; in these cases, the server applies its default values and policies. However, `username` e `password` are essential for each user. The field `confirm_password` is required by the API, but is generated automatically by the script and does not need to be supplied separately in the CLI or JSON file.

6.2 Successful responses

Depending on the operation, the utility treats status codes `200`, `201` e `204`. An existing group or user may be treated as an idempotent condition when the response `400` contains a duplication indication, como `exists`.

6.2 Group directions

The association is performed separately:

- `IN`: messages received by the user;
- `OUT`: messages sent by the user;
- `BOTH`: CLI shortcut that executes two requests, one for each direction.

For bidirectional tactical traffic, always validate `IN` e `OUT`.

7. CLI commands

7.1 Help

```
python ots_manager.py --help
python ots_manager.py create-user --help
python ots_manager.py qr --help
```

7.2 Create user e gerar QR Code Android

```
python ots_manager.py create-user \  
-u "piloto1" \  
-p "Senha123!" \  
-g CSAR \  
--app android
```

The command creates the group, creates the user, associates the group in the **IN** e **OUT**, requests the string from OTS, and saves a PNG with a default name such as **piloto1_android.png**.

7.3 Create user e gerar QR Code iPhone

```
python ots_manager.py create-user \  
-u "piloto2" \  
-p "Senha123!" \  
-g CSAR \  
--app iphone
```

7.4 Provide email and administrator profile

```
python ots_manager.py create-user \  
-u "operador" \  
-p "SenhaForte!" \  
-e "operador@empresa.com" \  
--admin \  
--app android
```

Use **--admin** somente quando a função operacional realmente exigir privilégios administrativos.

7.5 Validity and activation limit

```
python ots_manager.py create-user \  
-u "convidado" \  
-p "Senha123!" \  
-g Visitantes \  
--app android \  
--exp 30 \  
--max 1
```

--exp 30 represents 30 days from the time of generation. A date in the format can also be used: **YYYY-MM-DD**:

```
python ots_manager.py qr -u "piloto1" --app android --exp 2026-12-31 --max 2
```

When **--exp** e **--max** are omitted, these fields are not sent and the server applies its default policies.

7.6 Generate a QR Code for an existing user

```
python ots_manager.py qr -u "piloto1" --app android
python ots_manager.py qr -u "piloto2" --app iphone --save-qr piloto2_ios.png
```

7.7 Create group isolado

```
python ots_manager.py create-group -n "Patrulha"
```

7.8 Link a user to a group

```
python ots_manager.py link -u "piloto1" -g "Patrulha"
```

Specific direction:

```
python ots_manager.py link -u "piloto1" -g "Patrulha" --direction IN
python ots_manager.py link -u "piloto1" -g "Patrulha" --direction OUT
```

8. Batch provisioning

8.1 File `usuarios.json`

```
[
  {
    "username": "operador_alpha",
    "password": "SenhaForte123!",
    "email": "alpha@empresa.com",
    "administrator": false,
    "groups": ["CSAR", "Resgate"],
    "app": "android"
  },
  {
    "username": "operador_bravo",
    "password": "SenhaForte456!",
    "email": "bravo@empresa.com",
    "administrator": false,
    "groups": ["CSAR"],
    "app": "iphone"
  },
  {
    "username": "operador_temporario",
    "password": "TempPassword789!",
    "email": "temp@empresa.com",
    "administrator": false,
    "groups": ["Operacoes"],
    "app": "android",
    "expiration": 30,
    "max_uses": 1
  }
]
```

As chaves `expiration` e `max_uses` are optional. Se estiverem ausentes ou com valor `null`, as restrições não serão enviadas ao servidor.

8.2 Execution

```
python ots_manager.py batch \  
-f usuarios.json \  
-o resultado_qr_codes.json
```

Processing:

1. creates all groups found;
2. processes each user;
3. associates groups in `IN` e `OUT`;
4. converts validity to Unix Epoch when necessary;
5. requests the QR Code;
6. saves PNGs in `qr_codes/`;
7. exports the consolidated report.

8.3 Report structure

```
[  
  {  
    "username": "operador_alpha",  
    "app": "android",  
    "max_uses": "padrão do servidor",  
    "expiration": "padrão do servidor",  
    "qr_string": "string retornada pelo OTS",  
    "qr_image": "qr_codes/operador_alpha_android.png"  
  }  
]
```

The report must be treated as sensitive material. In real environments, consider generating an operational version without the `qr_string` key for sharing.

9. QR Codes and access validity

9.1 Android

The Android endpoint receives `username`. The optional fields are:

Field	Meaning
<code>exp</code>	expiration instant in Unix Epoch
<code>nbf</code>	instant from which the QR is valid
<code>max</code>	activation limit

When existe expiração, o script calcula `nbf` as the current instant in UTC.

9.2 iPhone

The iPhone endpoint is queried using `GET` and directly returns the iTAK configuration string. The response may be JSON, with keys such as `qr_string` ou `itak_qr_string`, or plain text.

9.3 Date conversion

A função `parse_expiration` aceita:

- number of days, such as `30`;
- absolute date, such as `2026-12-31`;
- empty values or values equivalent to `none`, `null`, `eterno` e `infinito`, treated as no restriction.

Absolute dates are interpreted in UTC. Confirm the timezone and server policy before using expiration dates in critical operations.

10. Operational procedures

10.1 Provision a new team

1. confirm the OTS address and operating account;
2. list the required groups;
3. review the JSON with another operator;
4. run the batch during an authorized window;
5. check the report and PNGs;
6. distribute each QR Code only to the correct recipient;
7. test login and CoT traffic on the device;
8. record the execution and validity date.

10.2 Reissue a QR Code

```
python ots_manager.py qr \  
--username "usuario_existente" \  
--app android \  
--save-qr "reemitido_usuario_android.png"
```

Confirm whether reissuing invalidates the previous QR Code according to OTS policy.

10.3 Validate post-provisioning connectivity

On the device:

- import the QR Code into the corresponding application;
- confirm the server address;
- confirm that the user can authenticate;
- test receipt of a CoT message;
- test sending a CoT message;
- validate that the expected groups are applied.

11. Error handling and diagnostics

Connection failure

Symptom: erro de conexão ou timeout.

Checks:

```
curl -i "$OTS_URL/"
getent hosts <hostname>
```

Confirm the IP, port, firewall, VPN, and that the OTS service is active.

Authentication failure

Symptom: mensagem `Falha na autenticação` com status HTTP diferente de `200` ou `201`.

Check the username, password, URL, login method, and whether the account is enabled.

HTTP 400 ao criar grupo ou usuário

This may indicate invalid data or duplication. O script reconhece duplicidade quando o corpo contém `exists`; otherwise, read the complete response and correct the submitted fields.

HTTP 401 ou 403 in a protected operation

Check:

- whether login actually created the session;
- whether the CSRF token exists in the cookies or response;
- whether the headers `X-CSRFToken` e `X-CSRF-TOKEN` were sent;
- se `Referer` e `Origin` match the `OTS_URL`;
- whether the account has permission for the operation;
- whether the session has expired.

Empty or invalid QR Code

Confirm the selected endpoint (`android` ou `iphone`), the user, HTTP status, and response format. Preserve the returned body during diagnostics without exposing tokens in public logs.

PNG not generated

Confirm the installation of `qrcode` e `pillow`, the existence of the destination directory and write permissions:

```
python -c "import qrcode, PIL; print('dependências OK')"
```

Partially completed batch

The batch may create some resources before failing on a later record. Do not rerun blindly. Compare the report, check which users and groups exist, and handle duplicates in a controlled manner.

12. Best practices and limitations

- Create a secure backup do arquivo de entrada antes de uma execução em lote.
- Use stable usernames and documented conventions.
- Avoid spaces and ambiguous characters in identifiers.
- Test first with a laboratory account.
- Do not distribute QR Codes through public channels.
- Record the generation date, validity, and person responsible for delivery.
- Always validate both group directions when there is bidirectional traffic.
- O modo `BOTH` executa duas requisições; uma falha pode deixar a associação incompleta.
- Handling “already exists” does not replace checking the current state on the server.
- The endpoint and response format may vary depending on the OpenTAKServer version.
- The original script does not implement transactional rollback. For critical batches, use smaller stages and subsequent reconciliation.
- The password appears on the command line in the examples; in production, prefer a secure input mechanism.

12.1 Notes de manutenção do código

When transcribing or updating the source code, validate the indentation of the block `get_qr_string`, especially the branches `elif app_type == "iphone"` e `else`. Run a syntax compilation before use:

```
python -m py_compile ots_manager.py
```

It is also recommended to add tests for `parse_expiration`, seleção dos endpoints, tratamento de respostas JSON/texto e montagem dos payloads.

13. Operations checklist

Before execution

- [] `OTS_URL` aponta para o ambiente correto.
- [] O acesso de rede foi validado.
- [] A conta tem permissão suficiente.
- [] A senha não está versionada.
- [] O JSON foi validado e revisado.
- [] Os nomes de grupos e usuários estão corretos.
- [] `app` é `android` ou `iphone`.
- [] Expiração e limite de uso foram conferidos.

After execution

- [] Login was successful.
- [] Groups were created or confirmed.
- [] Users were created or confirmed.
- [] Each group was associated in **IN** e **OUT**.
- [] The QR string was returned.
- [] The PNG was generated and opened for review.
- [] The JSON report was saved in a protected location.
- [] The device was tested in both communication directions.

14. Quick reference

```
# Instalação
pip install requests qrcode pillow

# Configuração
export OTS_URL="http://servidor-ots:5000"
export OTS_USER="admin"
export OTS_PASS="..."

# Usuário Android
python ots_manager.py create-user -u piloto1 -p 'Senha123!' -g CSAR --app android

# Usuário iPhone
python ots_manager.py create-user -u piloto2 -p 'Senha123!' -g CSAR --app iphone

# QR existente
python ots_manager.py qr -u piloto1 --app android --save-qr piloto1.png

# Grupo
python ots_manager.py create-group -n Patrulha

# Associação bidirecional
python ots_manager.py link -u piloto1 -g Patrulha

# Lote
python ots_manager.py batch -f usuarios.json -o resultado_qr_codes.json

# Validação sintática
python -m py_compile ots_manager.py
```

15. Appendix: code structure

O arquivo `ots_manager.py` está organizado pelas funções abaixo:

Function	Responsibility
<code>get_csrf_token()</code>	Retrieves the CSRF token from the session
<code>login()</code>	Authenticates and updates headers
<code>create_group()</code>	Creates or confirms a group
<code>create_user()</code>	Creates or confirms a user
<code>add_user_to_group()</code>	Associates groups in <code>IN</code> , <code>OUT</code> ou <code>BOTH</code>
<code>parse_expiration()</code>	Converts days/date to Unix Epoch
<code>get_qr_string()</code>	Gets the Android or iPhone configuration
<code>save_qr_code_image()</code>	Generates and saves the PNG image
<code>process_batch_list()</code>	Orchestrates batch provisioning
<code>main()</code>	Defines the CLI and dispatches commands

Conclusion

O OTS Manager reduces manual tasks and standardizes OpenTAKServer provisioning. Automation must be used together with access control, credential protection, input-file review, and functional validation on the ATAK/iTAK device.

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